ML Builder Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for ML Builder Expo before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone Train, Analyze, Improve

Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Model Cards and Human Review

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported ML Builder Expo trace and record the first step you can explain without help.

Use W2: Trace **evidence** for Machine learning project **defense**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed ML Builder Expo explanation against the supplied facts before discussing it.

Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the ML Builder Expo target to a trace and a justified conclusion.

Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied ML Builder Expo case and show the deciding evidence.

Final classifier gets16/20 correct; two of four errors are false negatives.

Report supported metrics and what is missing.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Your expo model succeeds on three demonstration plants. What **evidence** should accompany the demo and what **limitation** should you state?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 36 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what ML Builder Expo adds to Responsible Plant Classifier and keep the evidence.

Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Write an opening

> Write an opening. Apply the supplied ML Builder Expo facts and keep your reasoning.

Draft a three-sentence opening with purpose, a result and a limit.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace evidence

> Trace evidence. Apply the supplied ML Builder Expo facts and keep your reasoning.

Name the files needed to audit one reported accuracy figure.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Answer critique

> Answer critique. Apply the supplied ML Builder Expo facts and keep your reasoning.

A peer asks whether your model works for every school. Give a supported response.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied ML Builder Expo facts and keep your reasoning.

Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Plan the next step

> Plan the next step. Apply the supplied ML Builder Expo facts and keep your reasoning.

Write one next investigation tied to an observed **limitation**, and identify what must stay held out.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: ML Builder Expo

> Present a plant-classifier expo claim that another team could check. Explain the labelled step before checking the changed condition.

Present a plant-classifier expo claim that another team could check.

Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Present a plant-classifier expo claim that another team could check.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Present the three failure records and indoor-only **limitation** beside the headline rate.

1. Show failures

2. Show independence

3. Show denominator

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Show failures Present the three failure records and indoor-only **limitation** beside the headline rate. Read the picture in order.

Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The poster omits the baseline and calls the model “field ready”. Which conclusion is supported by the supplied facts?

1. The headline exceeds the supplied **evidence**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The headline exceeds the supplied **evidence**. No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Repair a missing building block

Review Capstone Train, Analyze, Improve (ai-4/35). Goal: Run a frozen final evaluation with complete accounting. Short practice: Why is “the revision is better” incomplete?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: What fresh **evidence** is needed after adding crossed **training** data?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, **reflection**s, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

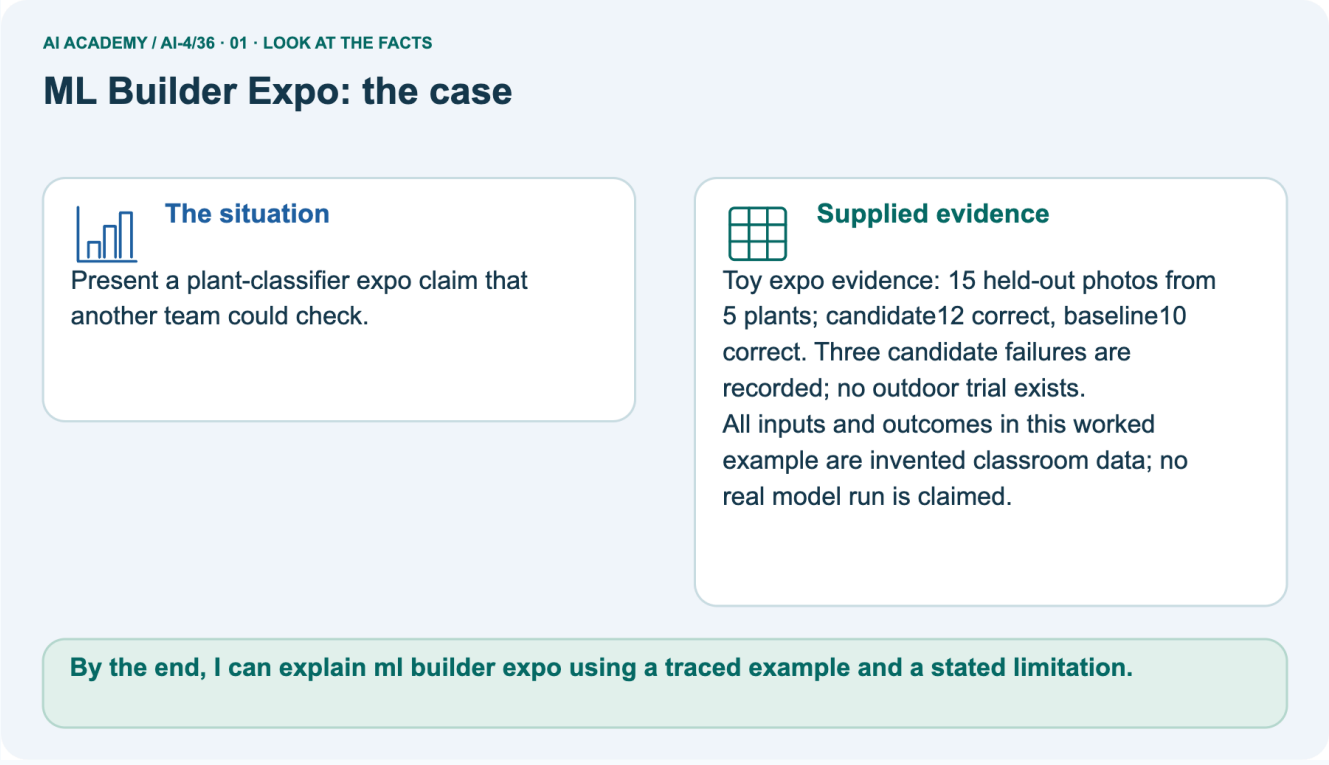
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each ML Builder Expo picture from its labels, then record a relevant limit.

01. ML Builder Expo: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

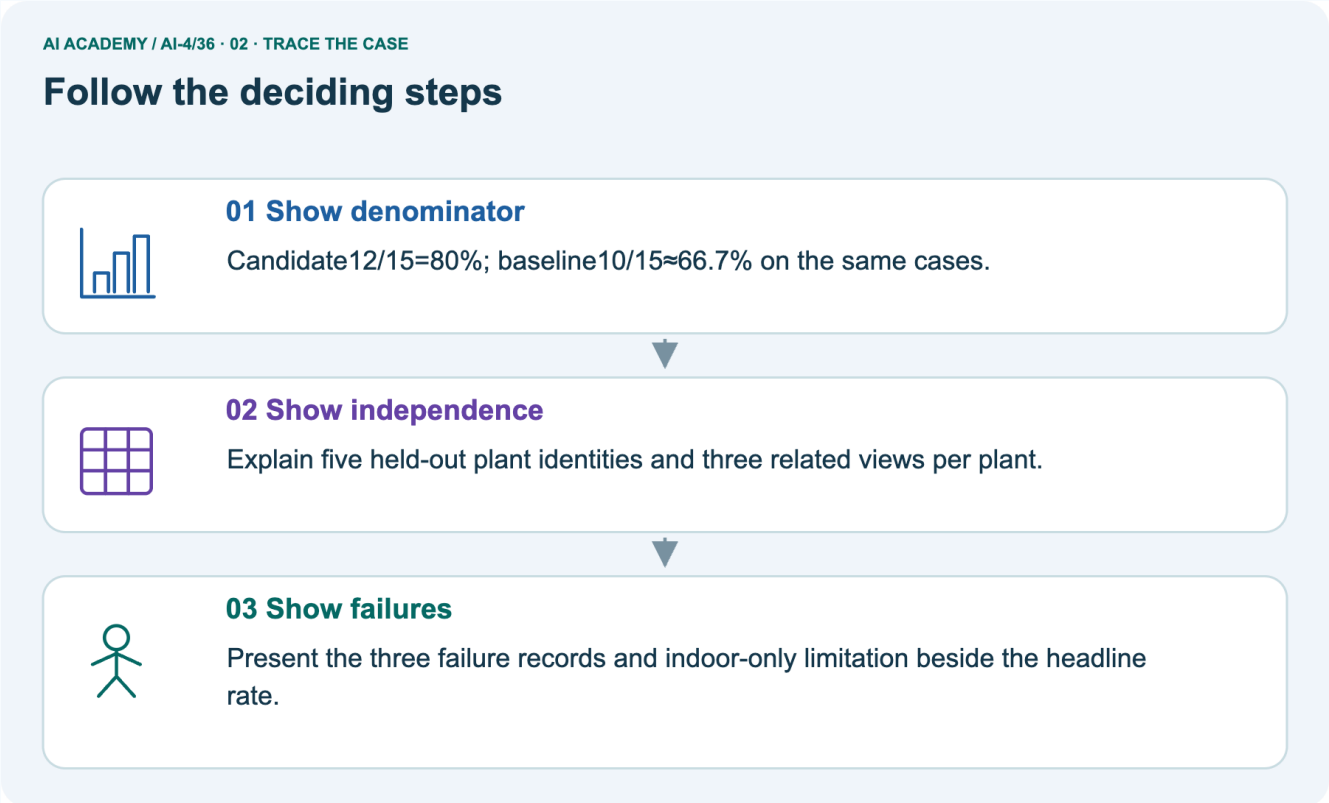
Present a plant-classifier expo claim that another team could check. Supplied facts: Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct.

Three candidate failures are recorded; no outdoor trial exists. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Show denominator: Candidate12/15=**80%**; baseline10/15≈66.7% on the same cases. → Show independence: Explain five held-out plant identities and three related views per plant. → Show failures: Present the three failure records and indoor-only **limitation** beside the headline rate.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Machine learning project defense at a glance

AI ACADEMY / AI-4/36 · 03 · INSPECT THE LESSON MODEL

Machine learning project defense at a glance

Part	Evidence or action	Interpretation
Show the process	Task data splits versions	Explain one prediction
Show the evidence	Baseline metrics errors	Trace one reported number
Show the limits	Unsupported uses next step	Answer peer questions

Prepare a five-minute presentation with a one-page evidence summary and linked project files. Include the data dictionary, split record, run record, prediction ledger, model card and revision history. Demonstrate one prediction or arithmetic calculation live so the audience can follow the reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

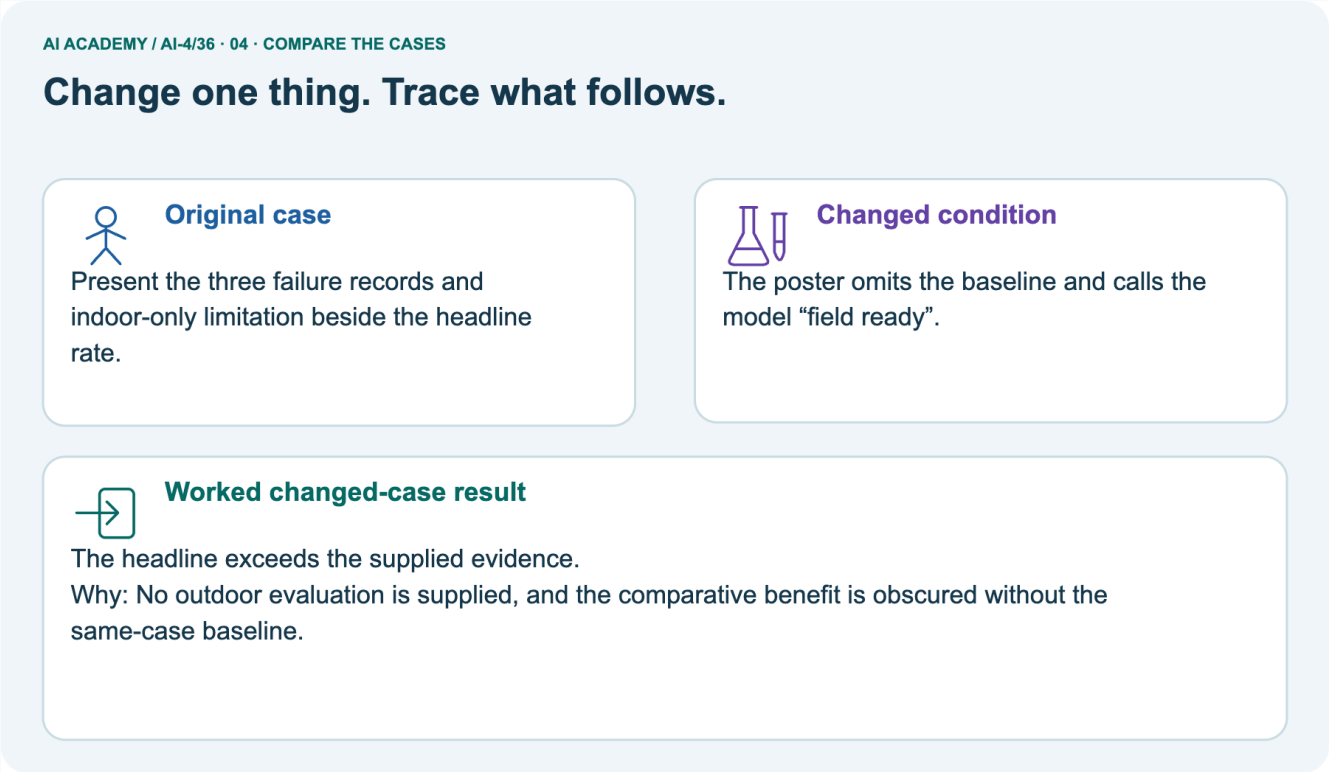
Machine learning project **defense** at a glance. Show the **process** | Task data splits versions | Explain one **prediction**. Show the **evidence** | Baseline metrics errors | Trace one reported number.

Show the limits | Unsupported uses next step | Answer peer questions

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

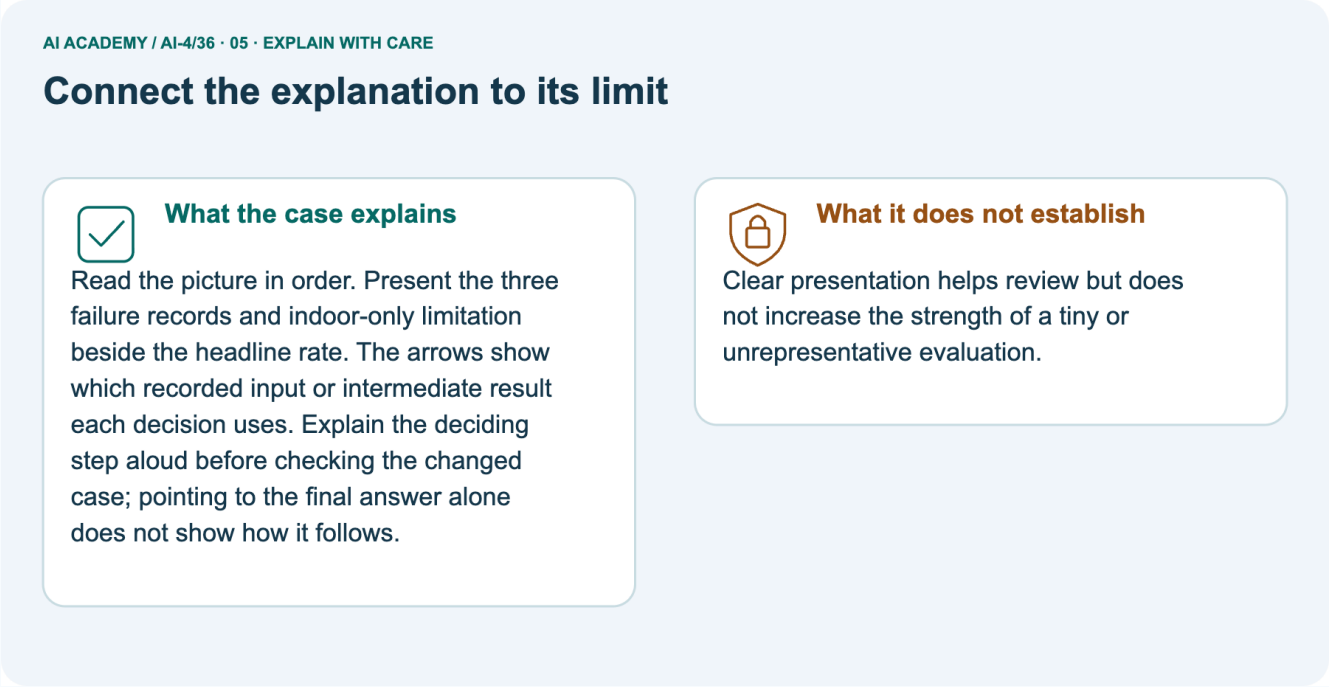
Original case: Present the three failure records and indoor-only **limitation** beside the headline rate.
Changed condition: The poster omits the baseline and calls the model “field ready”.

Expected result: The headline exceeds the supplied **evidence**. Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

> Explain the case, then point to a limit on the claim.

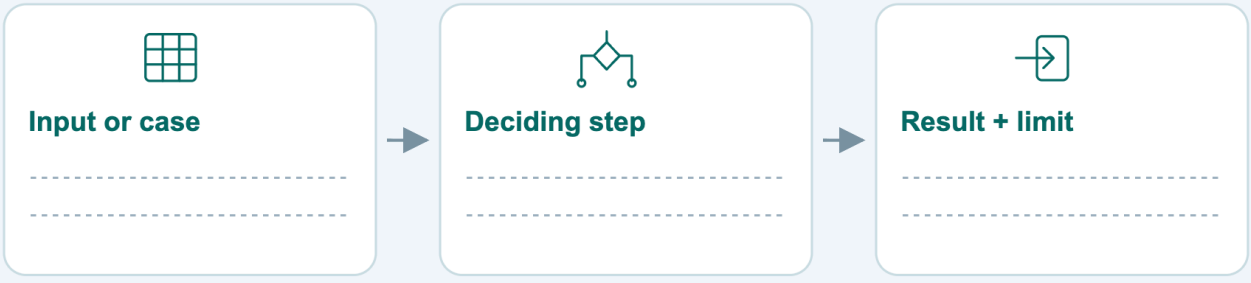
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/36 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.